

I built a customer
risk model.
Then I tested it on
the customers who
actually caused UK
banks to be fined.

Saman Barati · London · September 2026

Excel and Power Query. No machine learning, on purpose.

THE IDEA

Most risk models are judged on whether the maths works.

That is the easy test. A customer risk assessment can be perfectly consistent and still rate the wrong people Low.

So I built one properly — 20 risk factors mapped to regulation 18(2)(b) of the Money Laundering Regulations, 98 level definitions, every weight carrying a written rationale and a published source — and then asked a harder question.

If this model had been running on the day these customers were onboarded, would it have rated them High?

Six customers named in published FCA enforcement notices, rebuilt from what each bank knew at the time. No hindsight allowed in the inputs.

THE RESULT

£480m

of UK enforcement penalties between them.

All rated Low.

Nothing scored above 1.90 on a scale where High begins at 3.50. The single High rating came from a mandatory escalator, not from the score.

FINDING ONE

One word decided the most serious case.

Fowler Oldfield was a Bradford jewellery business. Around £365m passed through its NatWest account, roughly £264m of it in cash. It produced the FCA's first criminal prosecution of a bank.

Read as a cash-intensive trade	1.82 → Low
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Read as a dealer in high-value goods	1.90 → High
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Both readings are defensible on the wording I wrote. The scores differ by 0.075. The rating differs completely.

That is a defect in my level definitions, not in the analyst reading them.

FINDING TWO

A customer whose address was a London landmark scores 1.52.

Monzo onboarded customers who gave well-known London landmarks as their home address. My library has a factor written for exactly that — from that notice, before any of it was scored.

I then gave the factor **1.5% of the model**, and wrote a paragraph defending the choice.

Scored through, the customer lands comfortably in Low. The factor that saw the problem was too light to change the answer.

Both the reasoning and the result stay in the repository. Deleting the argument I got wrong would have removed the only evidence that the test worked.

FINDING THREE

The weights were never the thing that mattered.

I spent a whole document arguing about whether geography should carry 25% or 15%.

Move any category weight by 10 percentage points	up to 26 of 400 customers change band
The same change, against band boundaries placed where the population actually sits	29 to 100 customers change band

The boundaries decide the output. The weights barely touch it.

Which reverses the order of the work: fix the bands first, argue about the weights second.

The 2026 EDD change removes the only control catching a whole population.

The Money Laundering and Terrorist Financing (Amendment) Regulations 2026 narrow mandatory jurisdiction-based enhanced due diligence to the three FATF *Call for Action* countries, rather than those plus the 22 on the *Increased Monitoring* list.

Had jurisdiction-based EDD before 2026	16
Have it under the 2026 rule	5
Lose it	11
Of those, escalated for any other reason	0

The model still scores them. It just cannot move them across a band. The score sees the risk; the rating does not.

Five changes recommended. None applied.

- Move the band boundaries to where the population actually is
- Move “the declared profile cannot be true” out of the average and onto the escalator list
- Tie the high-value-dealer level to registration, not to a description of the trade
- Add a factor for how complete the file is — an unasked question currently scores the same as a reassuring answer
- Reduce the geography weight, once the bands are fixed

The methodology requires MLRO approval for any change to weights, bands or escalators. So they are written up and left there.

A model whose author quietly rewrites it the moment a test embarrasses him has no version history worth reading.

Synthetic data, a fictional bank, and a self-directed learning project alongside a BA in Accounting and Finance. Not professional advice, and not independently validated. Full method, workbooks and case files on GitHub — link in the comments.